UST-Net: A U-Shaped Transformer Network Using Shifted Windows for Hyperspectral Unmixing

Zhiru Yang, Mingming Xu, *Member, IEEE*, Shanwei Liu, Hui Sheng, and Jianhua Wan

***Abstract*— Autoencoders (AEs) are commonly utilized for acquiring low-dimensional data representations and performing data reconstruction, which makes them suitable for hyperspectral unmixing (HU). However, AE networks trained pixel by pixel and those employing localized convolutional filters disregard the global material distribution and distant interdependencies, resulting in the loss of necessary spatial feature information essential for the unmixing process. To overcome this limitation, we propose an innovative deep neural network model named U-shaped transformer network using shifted windows (UST-Net). UST-Net prioritizes spatial information in the scene that is more discriminative and significant by using multihead self-attention blocks based on shifted windows. Unlike patch-based unmixing networks, UST-Net operates on the complete image, eliminating inconsistencies associated with patches. Moreover, the downsampling and upsampling stages are used to extract hyperspectral image (HSI) feature maps at different scales. This process generates a context-rich and spatially accurate abundance map without losing local details. The experimental results of one synthetic dataset and three real datasets demonstrate that UST-Net significantly outperforms both traditional and several other advanced neural network methods. Our code is publicly available at https://github.com/UPCGIT/UST-Net.**

***Index Terms*— Autoencoder (AE), deep learning, hyperspectral unmixing (HU), multihead self-attention mechanism, transformer.**

I. Introduction

HYPERSPECTRAL images (HSIs) provide a comprehensive description of a scene by not only presenting its 2-D surface space but also providing spectral curves that correspond to different locations and objects. The high spectral resolution of HSIs enables the diagnosis of fine spectral features while identifying object types, materials, and material components. Therefore, HSIs are widely used in various fields, such as remote sensing research [1], food industry [2], [3], [4], biotechnology [5], medical science [6], agriculture [7], [8], and military defense [9].

Hyperspectral cameras have limited performance in terms of spatial resolution, resulting in low-quality HSIs. A single pixel usually covers tens or even thousands of meters, blurring the actual scene and producing mixed pixels. Mixed pixels are common in HSIs and pose a challenge to the development and application of HSIs. To solve this problem, hyperspectral unmixing (HU) is a critical technique that extracts components, referred to as endmembers, from HSI pixels, and calculates the percentage of each component, known as abundances. The significance of HU lies in its ability to extract important information from mixed pixels.

The linear mixing model (LMM) assumption is a widely used unmixing model that provides an acceptable approximation under the light-scattering regime in many scenarios [10]. In the traditional unmixing algorithm, endmember extraction is performed first, and then, the abundance is estimated using the LMM. The existing endmember extraction algorithms usually assume that endmembers are located at the corners or vertices of a linear simplex. The process of endmember extraction involves maximizing the volume of a closed simplex in hyperspectral data [11], [12], [13]. In the absence of pure pixels in the image, the endmembers are estimated by finding the smallest volume linear simplices containing the data points [14], [15], [16], [17]. Abundances are obtained by solving a linear model of an unknown variable, such as the least squares method [18]. However, the process of the traditional unmixing algorithm is prone to error accumulation. To avoid such errors, blind nonmixing techniques [19] for the simultaneous estimation of endmembers and abundances have gained great popularity. Existing approaches to study blind unmixing methods are generally based on nonnegative matrix factorization (NMF), which involves adding appropriate physical constraints during the matrix factorization process to estimate the endmembers and abundance simultaneously [20], [21], [22], [23], [24], [25].

In recent years, neural networks have been widely applied to HSI analysis tasks due to their remarkable general modeling capabilities engendered by the advancement of deep learning technology [26], [27], [28]. Two main types of neural networks are supervised and unsupervised networks. Supervised networks depend on labeled samples, which are both time-consuming and labor-intensive to prepare, and the accuracy of training is closely related to the quality of these labels. To overcome the limitations of supervised methods, unsupervised methods are considered as viable options, which are widely used in unmixing tasks. Among the unsupervised techniques, the autoencoder (AE) network approach, which is used for linear blind unmixing, has been widely studied. An AE consists of an encoder and a decoder, where the encoder maps the high-dimensional input to a low-dimensional embedding, and the decoder aims to reconstruct the original

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high-dimensional input from the low-dimensional embedding. This distinctive structure makes AEs suitable for unmixing tasks. Abundance can be viewed as a low-dimensional embedding of the HSI, with endmembers obtained during decoding, typically in the weights of the last hidden layer of the decoder. AE-based approaches have shown promising potential for exploiting the advanced denoising and intrinsic adaptive capabilities of neural networks. Furthermore, the structure of the AE can be variably designed to suit different unmixing tasks. By adding suitable constraints, meaningless unmixing results can be avoided, and the ability of the network to learn endmembers and abundance distributions can be significantly improved. In [29], a sparse decoupled denoising AE was proposed using denoising and the L_{21} constraint. In [30], an L_1-constrained sparse AE network was constructed. These constraints improve the robustness of the network. By designing different network loss functions, the network can also learn more reliable results. In [31], an AE with a Kullback–Leibler (KL) divergence term of spectral angular distance (SAD) similarity and an additional penalty term was proposed. In [32], a cycle-consistent unmixing network based on perceptual loss was proposed. The approach in [33] improves the effect of unmixing by introducing Wasserstein distance and feature matching as regularization terms and then adding them to the loss function. Regularized training of variational AE (VAE) also performs well in unmixing. VAE can solve the problem that two similar inputs in an AE are not necessarily encoded as points close together in latent space. A VAE was used in [34] to estimate abundance, and the VAE was used in [35] and [36] to learn a probability model of spectral variation and generate endmembers. In addition, multitask learning (MTL) is applied to HU. MTL can help the model achieve better generalization for the initial task by sharing representations between related tasks, thus enabling it to focus on crucial features. In [37], an MTL approach was used to implement a deep AE based on a bilinear mixed model. The two distinct tasks involve linear unmixing and updating bilinear components. The method in [38] used the attention mechanism to extract height features from light detection and ranging (LiDAR) data and utilized the height features to help with the HU task. Moreover, considering channel configurations and convolution kernels, the neural architecture search (NAS) was used to determine the optimal AE network structure [39]. However, these models are based on pixel learning and ignore the original spatial information of the HSI.

Several advanced deep neural networks that have emerged in the field of computer vision (CV) in recent years, such as long short-term memory (LSTM), generative adversarial network (GAN), convolutional neural network (CNN), transformer, and other state-of-the-art networks, have been adapted into AE structures and successfully applied to HU [40], [41], [42], [43], [44], [45], [46], [47], [48]. Simply using advanced networks as opaque mechanisms for unmixing only leads to reduced interpretability and limited generalization. Therefore, it is crucial to design and adjust the network precisely. In [41], an adversarial AE network was proposed, which used an adversarial training procedure to feed spatial information into the network. In [42] and [43], the convolutional layer was incorporated as the encoder's hidden layer, enabling the transformation of high-dimensional spatial representation into latent space representation. However, 2-D-CNN methods perform independent analysis across bands on the HSI, thereby ignoring the spectral nature of the data. In [44] and [45], a 3-D-CNN was designed as an encoder to learn the spatial structure and local spectral features of a given 3-D input image. However, it should be noted that the convolution operation extracts local features, and therefore, a considerable amount of contextual information inherent in the original HSI is lost. To overcome this problem, a convolutional and transformer-based network with multihead self-patch attention was proposed in [46] to capture nonlocal contextual feature correlations.

Previous research has demonstrated that improved outcomes can be attained through the incorporation of spatial information into the unmixing process. In fact, there are different scales and sizes of target objects, and pure pixels can appear distributed throughout the image, rather than being limited to local areas. However, due to the limited receptive field of the traditional CNN network, the network can only obtain local spatial information, making it difficult for the network to learn the overall distribution of features across the entire scene. To achieve accurate abundance estimation, it is crucial to consider the nonlocal spatial correlation between hyperspectral pixels and exploit the multiscale features of HSI [49]. The attention mechanism is considered as a feasible solution [46] because it can emphasize important information of interest while suppressing irrelevant contextual information. In numerous visual applications, the simplest way to enlarging the receptive field and attaining multiscale features involves employing downsampling and deeper models. However, in the context of HU, these techniques might face difficulty in reconciling the tradeoff between the loss of detailed information and the broader information acquired through downsampling.

In order to solve the above problems, we propose a U-shaped transformer network using shifted windows (UST-Net) for HU. This is an end-to-end network designed for extracting multiscale feature information from HSI. It incorporates a multihead self-attention block based on shifted windows, also known as the Swin transformer block (STB) [50]. The network efficiently captures the nonlocal spatial features of the HSI in an unsupervised manner to establish the long-range dependencies between pixels. While improving the accuracy of endmember extraction, a smoother abundance result can be obtained. More specifically, the main contributions of the proposed network in this article are summarized as follows.

1) An end-to-end unmixing network is proposed that captures the characteristics of HSI at different scales through the downsampling and upsampling stages. The network has a U-shaped structure similar to U-Net. This multiscale processing mechanism helps to better capture the details and structural information in the image. Consequently, the network's capability to comprehend information across different scales within the image is

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augmented, thereby leading to improved accuracy and stability in the unmixing process.

2) We formulate a module that utilizes the multihead self-attention block based on shifted windows, integrating it into both the downsampling and upsampling stages and ensuring the sharing of training weights across each subblock. This process encourages the network to focus on spatial information that is more discriminative and important in the scene while suppressing unnecessary contextual information. Thus, the AE encoder is guided to achieve more accurate abundance results.
3) Ablation and comparative experiments demonstrate the necessity of the network to gather multiscale information and confirm the superiority of the proposed module over traditional convolution filters. Furthermore, we evaluate the impact of different objective functions on the performance of the network.

The rest of this article is organized as follows. Section II briefly introduces the LMM and the related AE framework. Section III details the structure and details of the proposed network. Section IV presents the experimental methods and results. Finally, Section V concludes with some conclusions.

II. Related Works

Since the proposed method is based on LMM and AE framework, the formulation of this network will be introduced in Section II briefly.

A. Linear Mixing Model

The LMM indicates that each pixel in a spectral image is a result of combining linearly the spectra of various endmembers. Mathematically, this concept can be expressed as follows:

$$\boldsymbol{X} = \boldsymbol{W}\boldsymbol{S} + \boldsymbol{E} \tag{1}$$

where $\boldsymbol{X} \in \boldsymbol{R}^{L\times K}$ is the observed HSI matrix with L bands and K pixels, $\boldsymbol{W} = (\boldsymbol{W}_1, \boldsymbol{W}_2, \ldots, \boldsymbol{W}_p) \in \boldsymbol{R}^{L\times p}$ is the endmember spectrum library containing p endmembers, and $\boldsymbol{W}_i$ is the spectrum of L bands of the ith endmember. $\boldsymbol{S} \in \boldsymbol{R}^{p\times K}$ is the abundance matrix, and $\boldsymbol{E}$ is the noise matrix.

Moreover, both endmembers and abundances have distinct physical interpretations. The endmember represents the pure spectral signature of ground objects present in the HSI data. It must comply with the endmember nonnegativity constraint (ENC). The abundance refers to the relative contribution of each endmember within a pixel. It must adhere to both the abundance nonnegativity constraint (ANC) and the abundance sum-to-one constraint (ASC)

$$\boldsymbol{W}_i \geq 0 \tag{2}$$

$$\boldsymbol{S}_{ij} \geq 0, \quad \sum_{i=1}^{p} \boldsymbol{S}_{ij} = 1. \tag{3}$$

B. Autoencoder

AE networks are effective unsupervised learning models for unmixing tasks. Typically, an AE consists of two basic components: an encoder and a decoder. The encoder performs dimensionality reduction, while the decoder reconstructs data from the low-dimensional representation learned by the encoder.

1) Encoder: The encoder transforms the input data $\boldsymbol{X}$ into low-dimensional representations in the form of a set $\boldsymbol{S}$. The set $\boldsymbol{S}$ is composed of outputs from the encoder and is capable of representing the abundance. A three-layer encoder serves as an illustrative example. Its mathematical expression is given as follows:

$$\boldsymbol{S} = \sigma_3(\boldsymbol{W}_3\sigma_2(\boldsymbol{W}_2\sigma_1(\boldsymbol{W}_1\boldsymbol{X}))) \tag{4}$$

where σ_1–σ_3 represent the hidden layers of the encoder and $\boldsymbol{W}_1$–$\boldsymbol{W}_3$ represent the weights. No bias is added to the hidden layer, as during training, the bias tends to bring a considerably negative value [29]. The introduction of bias would also have an irreversible impact on the linear unmixing process. The final hidden layer of the encoder selects an activation function that generates nonnegative numbers, thus satisfying the ANC requirement for abundance.

2) Decoder: The decoder $g(\boldsymbol{S})$ is responsible for reconstructing the low-dimensional representation $\boldsymbol{S}$ to the original data. For instance, a one-layer linear decoder can be implemented using this mathematical expression

$$\hat{X} = g(\boldsymbol{S}) = \boldsymbol{W}\boldsymbol{S} \tag{5}$$

where the weight $\boldsymbol{W}$ connecting the hidden layer and the output layer in the decoder is denoted as the extracted endmember.

III. Proposed Method

The architecture of the proposed network is shown in Fig. 1. The encoder of the network extracts multiscale feature information and each subblock contains an STB to obtain the attention weight map.

A. Network Architecture

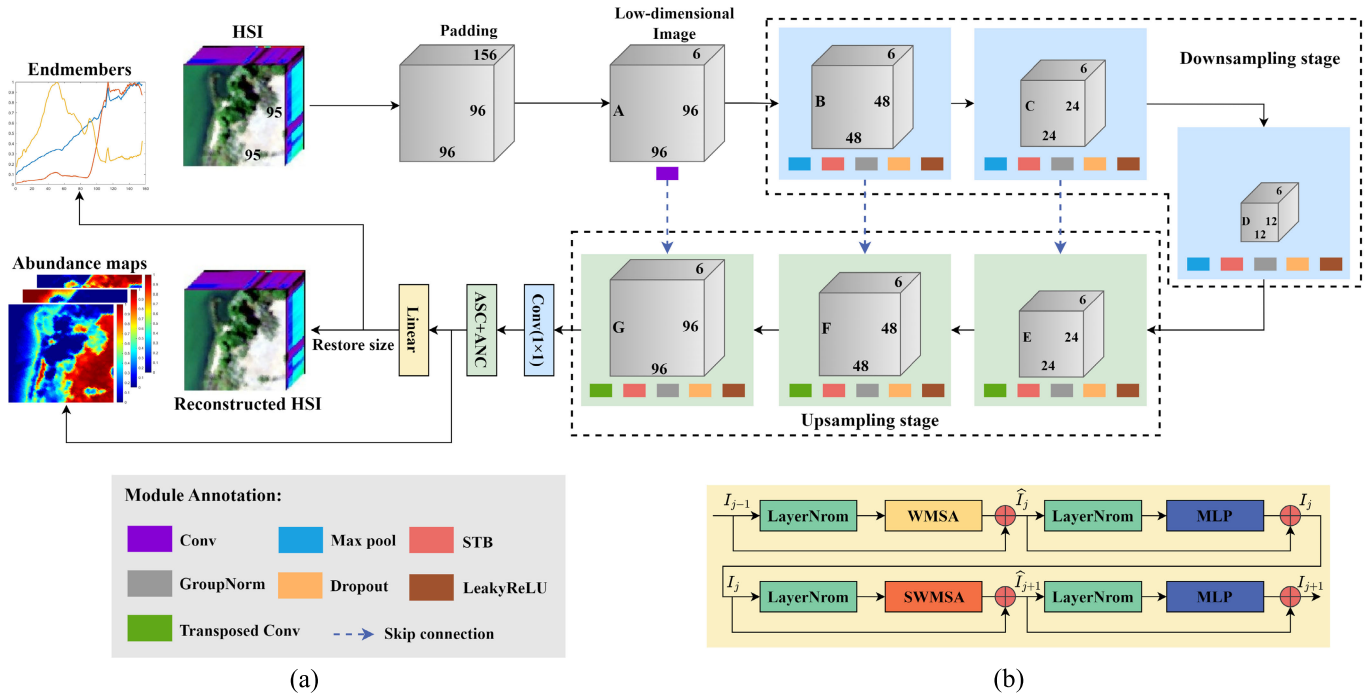
Prior to being fed into the network, the HSI $\boldsymbol{X}^{(L\times H\times W)}$ necessitates padding to make certain that rows H and columns W are divisible by 32. This is necessary since the network's downsampling stage conducts three 2×2 maximum pooling operations on the image, and the image needs to be divisible by the window size M (M is set to 4). Padding is to fill the pixels around the image with a value of 0. This padding procedure has no impact on subsequent training, as shown in Fig. 1, where $\boldsymbol{X}^{(156\times 95\times 95)}$ is padded as $\boldsymbol{X}^{(156\times 96\times 96)}$. Since the input HSI has many bands, a convolution layer that uses a convolution filter of 1×1 is implemented to reduce calculation while minimizing data dimensionality. Despite the fact that certain spectral information is lost, this stage enables the optimal low-dimensional representation, $\boldsymbol{A}^{(6\times 96\times 96)}$ of HSI to be attained during the network training process. Following this, $\boldsymbol{A}$ is input into the downsampling stage of the network, comprising a max-pooling layer, an STB, a GroupNorm layer, a dropout layer, and an activation layer in each of its subblocks. This process can be expressed as

$$\boldsymbol{B}_1 = \text{Max Pool}(\boldsymbol{A}) \tag{6}$$

$$\boldsymbol{B}_2 = \text{STB}(\boldsymbol{B}_1) \tag{7}$$

$$\boldsymbol{B}_3 = \text{GroupNorm}(\boldsymbol{B}_2) \tag{8}$$

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(a) (b)

Fig. 1. Architecture of the proposed UST-Net is designed for end-to-end training of the HSI (as shown in the figure, with a hyperspectral dataset Samson of 95 × 95 pixels and 156 bands used for demonstration). The downsampling stage is utilized to capture the extensive contextual information in the HSI, while the upsampling stage gradually restores the spatial resolution and minimizes information loss through skip connections. Each subblock in the stage comprises an STB. (a) Module annotation. (b) STB.

$$\boldsymbol{B}_4 = \text{Dropout}(\boldsymbol{B}_3) \tag{9}$$

$$\boldsymbol{B} = \text{LeakyReLU}(\boldsymbol{B}_4) = \begin{cases} \boldsymbol{B}_4 & \text{if } \boldsymbol{B}_4 > 0 \\ \alpha \boldsymbol{B}_4 & \text{if } \boldsymbol{B}_4 \leq 0 \end{cases} \tag{10}$$

where $\alpha = 0.3$. The resolution of the image output by the hidden layer is halved after passing through a max-pooling layer. Subsequently, the STB extracts the spatial characteristics of the image at this level, and dropout is applied to some neurons to prevent overfitting. Finally, the LeakyReLU function is applied to achieve nonlinear activation and accelerate network convergence. Every subblock can integrate data from nearby pixels and create direct connections with pixels that are far away. This process expands the receptive field, eventually yielding the deep-level spatial feature $\boldsymbol{B}^{(6\times48\times48)}$. Additional information regarding the STB and group normalization (GN) can be found in Sections III-B and III-C.

After three downsampling subblocks, the network produces image $\boldsymbol{D}^{(6\times12\times12)}$, which is then restored to its original resolution through the upsampling stage. Each upsampling subblock comprises the following subblocks in sequential order: a transposed convolutional layer with a 2 × 2 step size, an STB layer, a GroupNorm layer, a dropout layer, and an activation layer:

$$\boldsymbol{E}_1 = \text{Transposed Conv}\ (\boldsymbol{D}) \tag{11}$$

$$\boldsymbol{E}_2 = \text{STB}(\boldsymbol{E}_1 \oplus \boldsymbol{C}). \tag{12}$$

$$\boldsymbol{E}_3 = \text{GroupNorm}(\boldsymbol{E}_2) \tag{13}$$

$$\boldsymbol{E}_4 = \text{Dropout}(\boldsymbol{E}_3) \tag{14}$$

$$\boldsymbol{E} = \text{LeakyReLU}(\boldsymbol{E}_4) \tag{15}$$

After three upsampling subblocks, the image resolution is reconstructed. During this period, the output of the downsampling subblock is reinput into the upsampling subblock for concatenation, a technique known as skip connection or residual connection. As shown in Fig. 1, $\boldsymbol{D}$ enters the first upsampling block, and then, $\boldsymbol{E}_1$ is obtained through the transposed convolutional layer. The result $\boldsymbol{C}$ of the third downsampling subblock is fed into this block and connected in parallel with $\boldsymbol{E}_1$, such as in (12). Skip connections play a crucial role in reducing information loss and improving the reconstruction of individual pixels that gradually approach the original spatial distribution. After successfully executing three upsampling subblocks, $\boldsymbol{G}^{(6\times96\times96)}$ is obtained, which acts as a low-dimensional feature representation of $\boldsymbol{X}$ and represents an optimized version of $\boldsymbol{A}$. Finally, $\boldsymbol{G}$ goes through a 1 × 1 convolution layer and a softmax layer resulting in $\boldsymbol{S}^{(3\times96\times96)}$

$$\boldsymbol{H} = \text{Conv}(\boldsymbol{G}) \tag{16}$$

$$\boldsymbol{S} = \text{Softmax}(\gamma \boldsymbol{H}) \tag{17}$$

where γ is a constant greater than 1, the scaled softmax function considers both ASC and ANC, and when a higher value of γ is chosen, it has a similar effect to $L1$-norm regularization on the abundance map [48]. Subsequently, the output $\boldsymbol{S}$ of this hidden layer is the abundance obtained by network training. Finally, the HSI is reconstructed through a linear fully connected layer, and the reconstructed HSI is obtained by removing the filled pixels to restore the image size. Moreover, the linear layer's weight represents the network-trained endmember $\boldsymbol{W}$

$$\hat{X} = \text{Linear}(\boldsymbol{S}) = \boldsymbol{W}\boldsymbol{S}. \tag{18}$$

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B. Multihead Self-Attention Block Based on Shifted Windows

The STB utilizes the standard multihead self-attention mechanism of the original Transformer architecture [51]. The difference of STB is that it combines a local attention mechanism with a window shifting mechanism to achieve a larger attentional area and reduce computational complexity. The image preprocessing part of the Swin transformer is removed [50], including patch embedding and patch merging. Then, incorporate the extracted STB into our network to obtain the self-attention of pixels in a single $M \times M$ window (i.e., local attention) and self-attention between different windows (i.e., global attention) without patching the image.

For a local window feature $\boldsymbol{I} \in \boldsymbol{R}^{M^2 \times C}$, where C is the number of channels of this feature, we set the window size $M = 4$, and the query, key, and value matrices $\boldsymbol{Q}$, $\boldsymbol{K}$, and $\boldsymbol{V}$ are computed as

$$\boldsymbol{Q} = \boldsymbol{I}\boldsymbol{P}_Q, \boldsymbol{K} = \boldsymbol{I}\boldsymbol{P}_K, \boldsymbol{V} = \boldsymbol{I}\boldsymbol{P}_V \tag{19}$$

where $\boldsymbol{P}_Q, , \boldsymbol{P}_K, \boldsymbol{P}_V \in \boldsymbol{R}^{C \times f}$ are projection matrices shared between different windows and $\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V} \in \boldsymbol{R}^{M^2 \times f}$, with f the channel number of projected features. Therefore, the attention matrix is computed by the self-attention mechanism in the local window as

$$\text{Attention}(\boldsymbol{Q}, \boldsymbol{K}, \boldsymbol{V}) = \text{Softmax}\left(\frac{\boldsymbol{Q}\boldsymbol{K}^T}{\sqrt{f}} + \boldsymbol{B}\right)\boldsymbol{V} \tag{20}$$

where $\boldsymbol{B}$ represents a relative positional encoding that is learnable. The attention function is executed h times simultaneously to obtain window-based multihead self-attention (WMSA), where h denotes the number of heads in multihead self-attention, and generally, $f = C/h$. Subsequently, a multilayer perceptron (MLP) using two fully connected layers along with a Gaussian error linear unit (GELU) nonlinearity is employed to transform the features. Prior to multi-head self attention (MSA) and MLP, a layer normalization layer is added, and both modules leverage residual connections. As shown in Fig. 1(b), the whole process is formulated as

$$\hat{\boldsymbol{I}}_j = WMSA(LayerNorm(\boldsymbol{I}_{j-1})) + \boldsymbol{I}_{j-1} \tag{21}$$

$$\boldsymbol{I}_j = MLP(LayerNorm(\hat{\boldsymbol{I}}_j)) + \hat{\boldsymbol{I}}_j. \tag{22}$$

Then, the window is shifted by $(\lfloor (M/2) \rfloor, \lfloor (M/2) \rfloor)$ pixels from the regularly divided window, and the attention matrix of the shifted WMSA (SWMSA) is calculated to realize the information interaction between windows. The whole process is formulated as

$$\hat{\boldsymbol{I}}_{j+1} = SWMSA(LayerNorm(\boldsymbol{I}_j)) + \boldsymbol{I}_j \tag{23}$$

$$\boldsymbol{I}_{j+1} = MLP(LayerNorm(\hat{\boldsymbol{I}}_{j+1})) + \hat{\boldsymbol{I}}_{j+1}. \tag{24}$$

C. Group Normalization

As UST-Net engages in training over the entire HSI, it commonly uses a small batch size, often as small as 1. However, the commonly used normalization method batch normalization (BN) degrades the network performance under this circumstance. GN is still normalized in essence, but the normalization method of GN avoids the influence of batch size on the model. When GN calculates the mean and standard deviation, it divides the channel of each sample feature map into V groups, with each group having C/V channels. Then, it calculates the mean and standard of the elements in these channels' differences. Each set of channels is independently normalized with its corresponding normalization parameter. When V is equal to C and 1, GN corresponds to instance normalization and layer normalization, respectively. The specific calculation formula is given as follows:

$$\mu_i = \frac{1}{\left(\frac{C}{V}\right)HW} \sum_{c=\frac{iC}{V}}^{\frac{(i+1)c}{V}} \sum_{h=1}^{H} \sum_{w=1}^{W} x_{(n,c,h,w)} \tag{25}$$

$$\sigma_i^2 = \frac{1}{\left(\frac{C}{V}\right)HW} \sum_{c=\frac{iC}{V}}^{\frac{(i+1)C}{V}} \sum_{h=1}^{H} \sum_{w=1}^{W} \left(x_{(n,c,h,w)} - \mu_i\right)^2 \tag{26}$$

$$\hat{x}_i = \frac{x_i - \mu_i}{\sqrt{\sigma_i^2 + \epsilon}} \tag{27}$$

where x is the feature computed by a layer and i is the index. Formally, μ and σ represent standard deviation and variance, respectively. Also, V is the number of groups, w represents a predefined hyperparameter, and C/V is the number of channels per group.

D. Loss Function

The neural network objective function uses either mean square error (MSE) or SAD as the metric matrix. Although MSE can accurately reconstruct the image, it is vulnerable to the spectrum's scale, which can result in high reconstruction errors during training. On the other hand, SAD is a scale-invariant objective function that can effectively restore the feature space distribution of images. To achieve optimal performance and accelerate the convergence of the entire model, the total loss function is designed as the average weighted sum of these two losses

$$L_{\text{MSE}}(\mathbf{X}, \widehat{\mathbf{X}}) = \frac{1}{K} \sum_{i=1}^{K} \left(\widehat{\mathbf{X}}_i - \mathbf{X}_i\right)^2 \tag{28}$$

$$L_{\text{SAD}}(\mathbf{X}, \widehat{\mathbf{X}}) = \frac{1}{K} \sum_{i=1}^{K} \arccos\left(\frac{\left\langle \mathbf{X}_i, \widehat{\mathbf{X}}_i \right\rangle}{\|\mathbf{X}_i\|_2 \left\|\widehat{\mathbf{X}}_i\right\|_2}\right) \tag{29}$$

$$\text{Loss}_{\text{total}} = L_{\text{MSE}} + L_{\text{SAD}}. \tag{30}$$

IV. Experiments

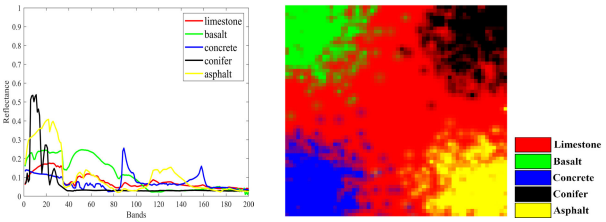
To evaluate the effectiveness of the proposed network, we perform experiments using one synthetic dataset and three real hyperspectral datasets, subsequently comparing it with the existing methods. The following are the unmixing methods and networks studied: vertex component analysis (VCA) [12] and fully constrained least squares (FCLS) [18] are traditional geometry-based methods; spatial group sparsity-regularized NMF [20] is an NMF method that considers spatial information; untied denoising AE with sparsity (uDAS) [29] and endmember extraction network (EndNet) [31] are networks that focus on spectral information; and cycle-consistency unmixing network (CyCU-Net) [32], convolutional AE unmixing network (CNNAEU) [43], and deep transformer network (DeepTrans) [46] are networks prioritizing spatial information.

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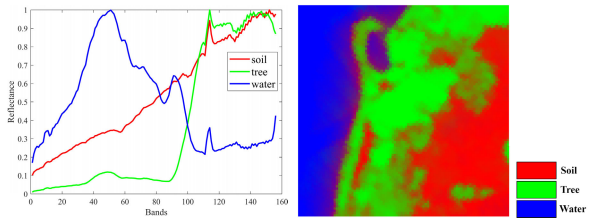
(a) (b) (c)

Fig. 2. Real data used in the experiments. (a) Samson. (b) Jasper Ridge. (c) Washington DC Mall.



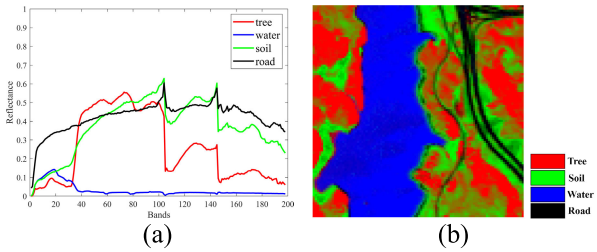
(a) (b)

Fig. 3. (a) Five pure object spectra used in the simulated data. (b) Abundances of simulated data.



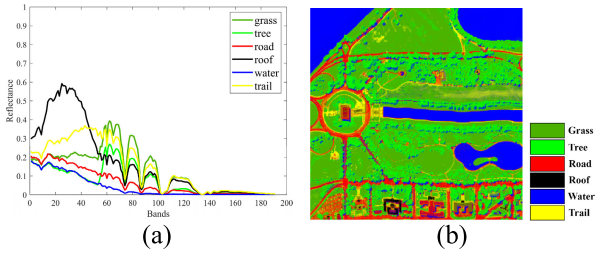
(a) (b)

Fig. 4. (a) Endmembers of Samson dataset. (b) Abundances of Samson data.



(a) (b)

Fig. 5. (a) Endmembers of Jasper dataset. (b) Abundances of Jasper data.



(a) (b)

Fig. 6. (a) Endmembers of Washington DC Mall dataset. (b) Abundances of Washington DC Mall data.

A. Datasets

The real datasets are shown in Fig. 2, and the detailed data are introduced in the following.

1) *Simulated Dataset:* Five spectra are randomly selected from the ASTER spectral library as endmembers, and the number of bands is 200. Endmembers are shown in Fig. 3(a); they are "#1 limestone," "#2 basalt," "#3 concrete," "#4 conifer," and "#5 asphalt." To add spatial context, we employ the abundance distribution detailed in [52], as shown in Fig. 3(b). Finally, the noise is added to the generated mixed data, and the signal-to-noise ratios (SNRs) are 10, 20, and 30 dB.
2) *Samson Dataset*: The original data have 952 × 952 pixels and 156 channels. Each pixel is recorded on 156 channels covering wavelengths from 401 to 889 nm. The spectral resolution is as high as 3.13 nm. Since the original image is too large, this is computationally expensive, so a region of 95 × 95 pixels is used. These data are not degraded by blank or heavily noisy channels. Specifically, there are three feature endmembers in this image, as shown in Fig. 4, which are "#1 soil," "#2 tree," and "#3 water." These endmembers are chosen within the image and closely resemble the ground spectrum. Therefore, these reference endmembers are regarded as ground truth (GT).
3) *Jasper Ridge Dataset*: The raw data have 512 × 614 pixels. Each pixel is recorded in 224 channels ranging from 380 to 2500 nm. The spectral resolution is as high as 9.46 nm. Since this HSI is too complex to get GT, we consider subimages of 100 × 100 pixels. Following the exclusion of channels 1–3, 108–112, 154–166, and 220–224 to account for dense water vapor and atmospheric effects, a total of 198 channels are retained. The endmembers are extracted from the image and considered as GT, as shown in Fig. 5(a).
4) *WashingtonDC Mall*: This HSI was collected over a Washington DC mall using the hyperspectral digital image acquisition experiment (HYDICE) sensor [53]. The original data have 1280 × 307 pixels. The cropped image is an area of 290 × 290 pixels. The data have a total of 191 bands, covering wavelengths from 400 to 2400 nm. Four pure matter endmembers are obtained from this image [46], namely, "#1 grass," "#2 tree," "#3 roof," "#4 road," "#5 water," and "#6 trail." Also, the abundance images are generated from endmembers using FCLS, as shown in Fig. 6(b).

B. Experimental Setup

1) Network Details Settings: The number of attention heads is an adjustable variable known as a hyperparameter, h. In addition, the number of filters in the initial convolutional

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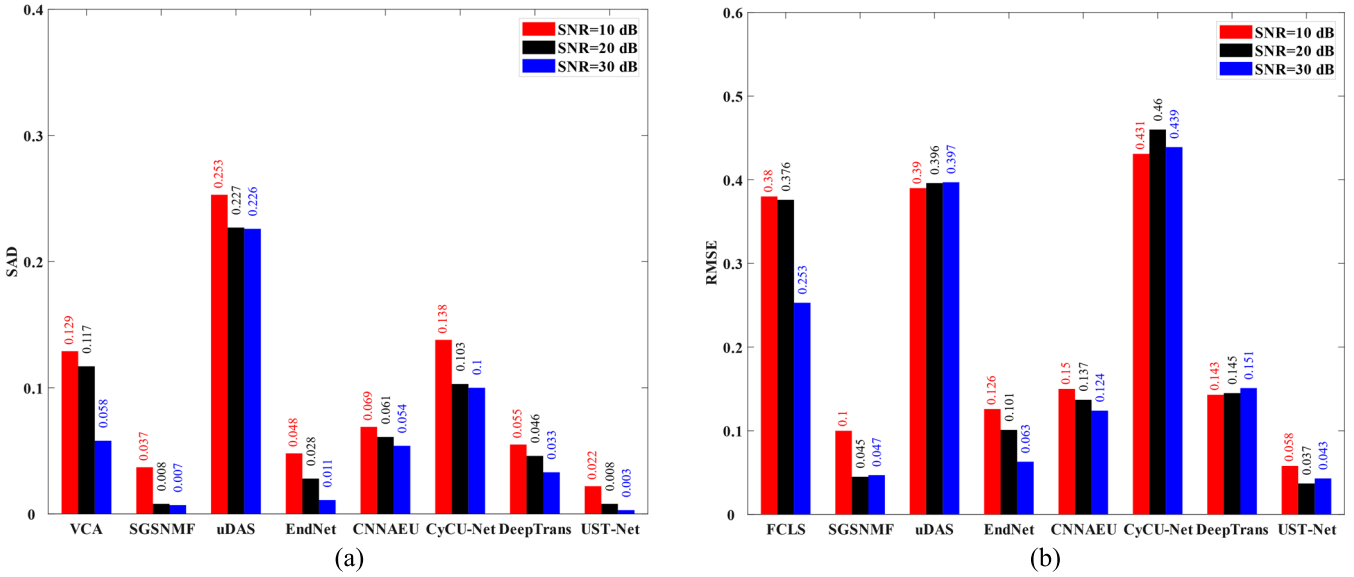


Fig. 7. Robustness evaluation of competing methods and UST-Net on simulated data with different SNRs. (a) SAD. (b) RMSE.

TABLE I
NETWORK PARAMETER SETTINGS

Parameter	Simulated data	Samson	Jasper	Washington DC Mall
h	6	6	5	5
d	18	6	15	15
γ	3	1	1	3
learning rate	0.001	0.001	0.001	0.006
epoch	200	100	100	100

layer is also a variable hyperparameter, which is denoted by d and represents the dimension of $\boldsymbol{A}$ in Fig. 1. To ensure uniform computational load across each head, d is set to be a multiple of h. The number of filters in the second convolutional layer is equal to the number of endmembers. Moreover, the number of groups V of the GN layer in the network is equal to the number of attention heads h, i.e., $V = h$. A fraction of neurons is randomly dropped in each dropout layer, and the rate is set at 20%. The scale of softmax, represented by the hyperparameter γ, should also be considered. The model is trained by stacking ten HSIs into the training data input network with a maximum number of epochs equal to 100. The network optimization strategy employed is RMSprop, and the network is trained with random initialization. The hyperparameters for different datasets are tabulated in Table I.

2) Quantitative Evaluation Indicators: Two indicators, namely, the SAD of the endmember and the root MSE (RMSE) of the abundance, are utilized to evaluate the algorithm's unmixing effect. The following formulas define these indicators:

$$\mathrm{SAD} = \frac{1}{p}\sum_{i=1}^{p}\arccos\left(\frac{\langle \widehat{\boldsymbol{W}}_i, \boldsymbol{W}_i\rangle}{\|\widehat{\boldsymbol{W}}_i\|_2\|\boldsymbol{W}_i\|_2}\right) \tag{31}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{p*K}\sum_{j=1}^{K}\|\widehat{\boldsymbol{S}}_j - \boldsymbol{S}_j\|_2^2} \tag{32}$$

where $\hat{W}_i$ and $\boldsymbol{W}_i$ denote the extracted endmember and the library spectrum, respectively, and $\boldsymbol{S}_j$ and $\hat{S}_j$ denote the abundance GT and the estimated values, respectively.

C. Network Noise Robustness Experiment

In order to evaluate the network's robustness to noise, simulated datasets are designed with different SNRs. The performance of the network is evaluated using two metrics, SAD and RMSE, as shown in Fig. 7. Our proposed network outperforms other state-of-the-art methods on datasets with SNRs of 10, 20, and 30 dB, demonstrating its strong performance. Generally, as the noise level increases, the unmixing performance of the algorithm worsens. However, there are exceptions to this trend. SGSNMF performs well on simulated data with higher SNR, but its unmixing performance decreases at high noise levels, which indicates that the algorithm is less robust to noise. The simulated data emulate the distribution of ground spatial information. Among all the networks designed to utilize spatial information, the UST-Net shows exceptional learning ability for multiscale advanced semantic information of data. Moreover, the performance of UST-Net does not degrade drastically as the noise level increases, indicating its good noise robustness. To demonstrate the effectiveness of our proposed network, we compare the endmember extraction results and the abundance extraction results with those of the real value on data with an SNR of 30 dB, as shown in Fig. 8. The results indicate that both the proposed endmember and abundance are highly similar to the real value.

D. Experiment With Real DataSets

1) Samson: A table with a quantitative overview of the performance evaluation of the methods can be found in Table II,

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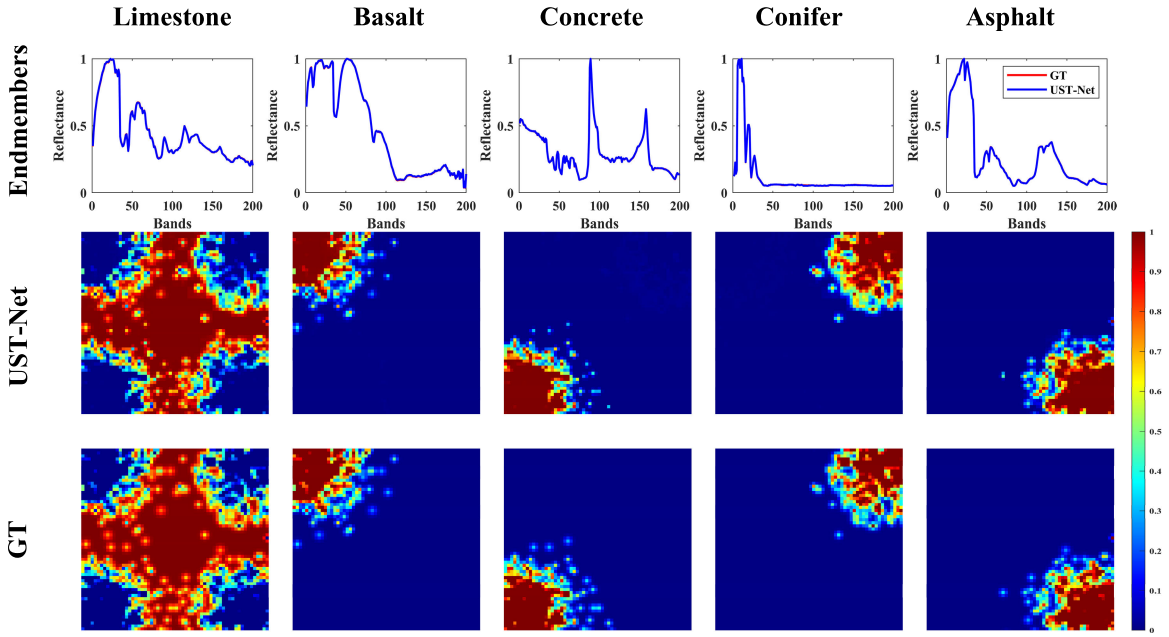


Fig. 8. Endmember extraction and abundance estimation results of the simulated data (30 dB).

TABLE II

QUANTITATIVE COMPARISONS ON THE SAMSON DATASET. THE BEST RESULTS ARE SHOWN IN RED, AND THE SUBOPTIMAL RESULTS ARE SHOWN IN BLUE

Methods		VCA-FCLS	SGSNMF	uDAS	EndNet	CyCU-Net	CNNAEU	DeepTrans	UST-Net
SAD	Soil	0.0236	0.0099	0.0300	0.0174	0.0106	0.0482	0.0085	0.0091
	Tree	0.0399	0.0450	0.0679	0.0343	0.0250	0.0413	0.0341	0.0332
	Water	0.1504	0.2290	0.1668	0.0501	0.0555	0.1213	0.0526	0.0276
Mean SAD		0.0713	0.0946	0.0882	0.0339	0.0303	0.0703	0.0317	0.0233
RMSE		0.3243	0.2692	0.2977	0.0648	0.1777	0.2372	0.1562	0.0645

with the corresponding endmembers and abundance results shown in Figs. 9 and 10. Although the Samson dataset is relatively straightforward, with a visibly defined ground feature distribution, some methods still fail to deliver sufficient outcomes. Traditional methods, NMF-based methods, and uDAS show the weakest results due to their reliance on VCA-based initialization methods that produce limited extraction effects that limit the method's performance. Even though EndNet, CyCU-Net, and DeepTrans are all initialized with VCA and perform better than the three previously mentioned methods, their performance remains inferior to the proposed method. As for the proposed method, it provides optimal results in both SAD and RMSE, with endmember and abundance results closest to the true values.

2) Jasper: Table III provides a quantitative summary of the performance evaluations of all algorithms. Figs. 11 and 12 show the endmember and abundance results, respectively. The Jasper Ridge dataset has greater spatial complexity and contextual information compared to the Samson dataset. The proposed method delivers excellent results, as can be observed from Table III in terms of SAD and RMSE. Fig. 11 shows that endmembers, which the network extracts, maintain consistency with the actual value. Notably, Fig. 12 shows the effectiveness of a spatial information-driven network (except CNNAEU) in surpassing networks that rely on spectral information such as FCLS, uDAS, and EndNet. In addition, the proposed method surpasses other spatial information-aligned networks such as CyCU-Net, CNNAEU, and DeepTrans in terms of local detail abundance and overall coherency.

3) Washington DC Mall: The performance evaluations of all algorithms on the Washington DC Mall dataset are shown in Table IV, with endmember and abundance results shown in Figs. 13 and 14, respectively. Compared to the Samson and Jasper Ridge datasets, the Washington DC Mall dataset is significantly larger in size and has a more complex feature configuration. Table IV shows that obtaining a good endmember set is relatively challenging. Nevertheless, our proposed method achieves the optimal outcomes. The UST-Net is unable to accurately identify the abundance of the roof in Fig. 14 because roofs only occupy a small portion of the entire scene. However, the proposed network shows the closest agreement with the GT for overall abundance results. This accuracy is evidence of the effectiveness of the proposed network in developing an accurate representation of the actual layout of the abundance.

E. Ablation Experiment

Experiments with different scales are performed to validate the effectiveness of the multiscale framework in HU tasks. We train six groups of networks, including: 1) AE network

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TABLE III

QUANTITATIVE COMPARISONS ON THE JASPER RIDGE DATASET. THE BEST RESULTS ARE SHOWN IN RED, AND THE SUBOPTIMAL RESULTS ARE SHOWN IN BLUE

Methods		VCA-FCLS	SGSNMF	uDAS	EndNet	CyCU-Net	CNNAEU	DeepTrans	UST-Net
SAD	Tree	0.1481	0.1291	0.1604	0.0277	0.0324	0.0730	0.0468	0.0639
	Water	0.1056	0.1943	0.0765	0.0417	0.1536	0.0542	0.0314	0.0292
	Soil	0.1166	0.1239	0.1130	0.0286	0.0323	0.1533	0.0808	0.0373
	Road	0.0901	0.0505	0.0595	0.5903	0.0401	0.1030	0.0864	0.0425
Mean SAD		0.1151	0.1245	0.1024	0.1721	0.0646	0.0959	0.0614	0.0432
RMSE		0.1569	0.1340	0.1576	0.2008	0.1153	0.3130	0.1210	0.0726

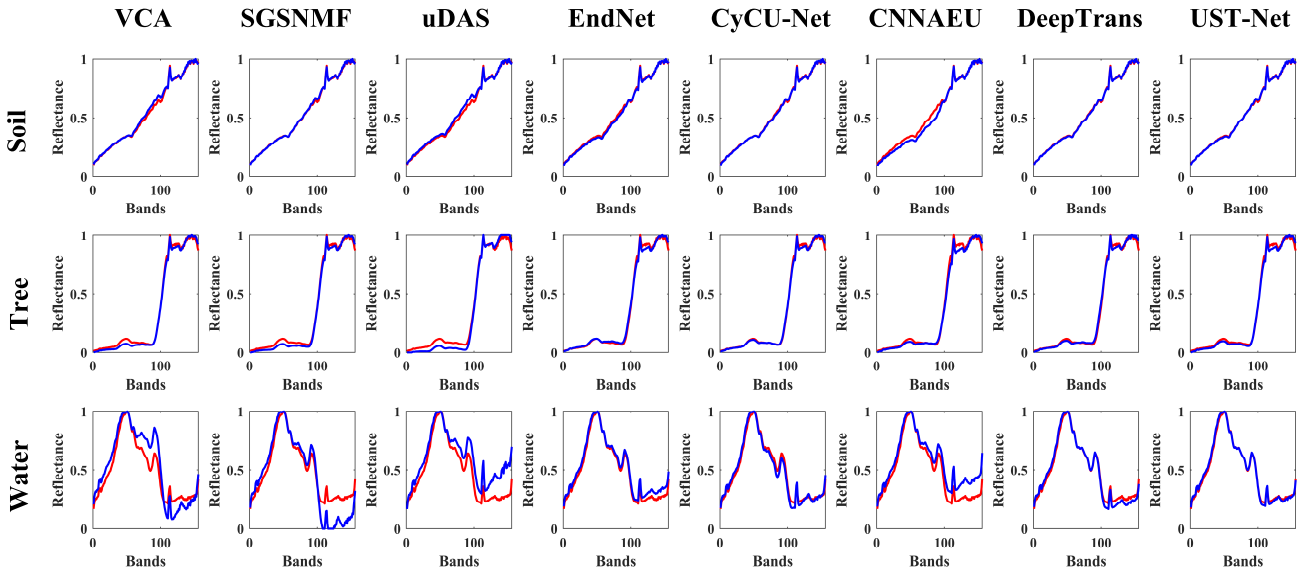


Fig. 9. Endmember extraction results of the Samson dataset, where the red line is the true value and the blue line is the extraction results of different methods.

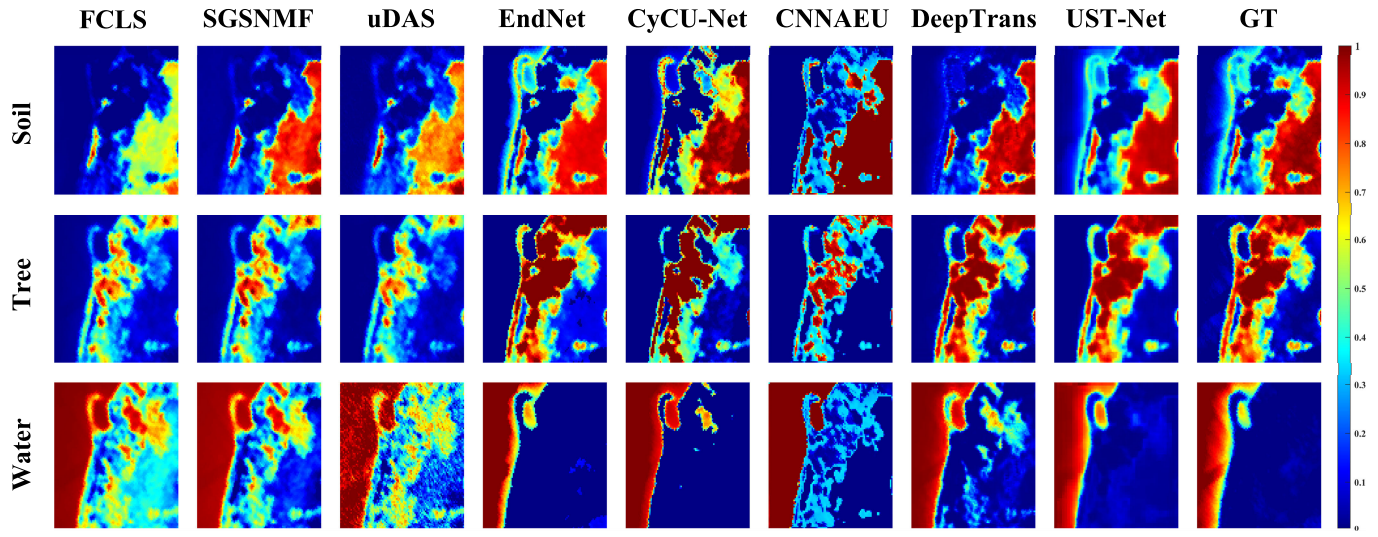


Fig. 10. Comparison of the abundance estimation results of soil, tree, and water with the abundance GT.

based on STB with one downsampled subblock and upsampled subblock (down-up block); 2) AE network based on STB with two down-up blocks; 3) AE network based on STB with three down-up blocks; 4) AE network based on 3 × 3 convolution filter with three down-up blocks; 5) AE network based on 5 × 5 convolution filter with three down-up blocks; and 6) AE network based on 7 × 7 convolution filter with three down-up blocks. The experimental results are shown in Table V.

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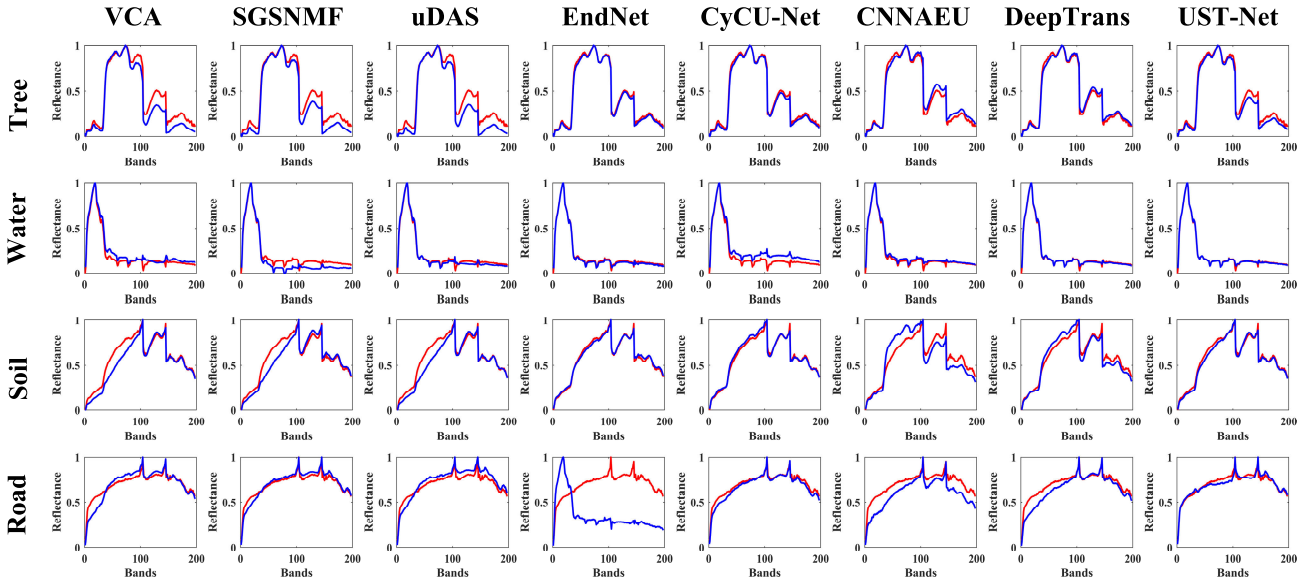


Fig. 11. Endmember extraction results of the Jasper Ridge dataset, where the red line is the true value and the blue line is the extraction results of different methods.

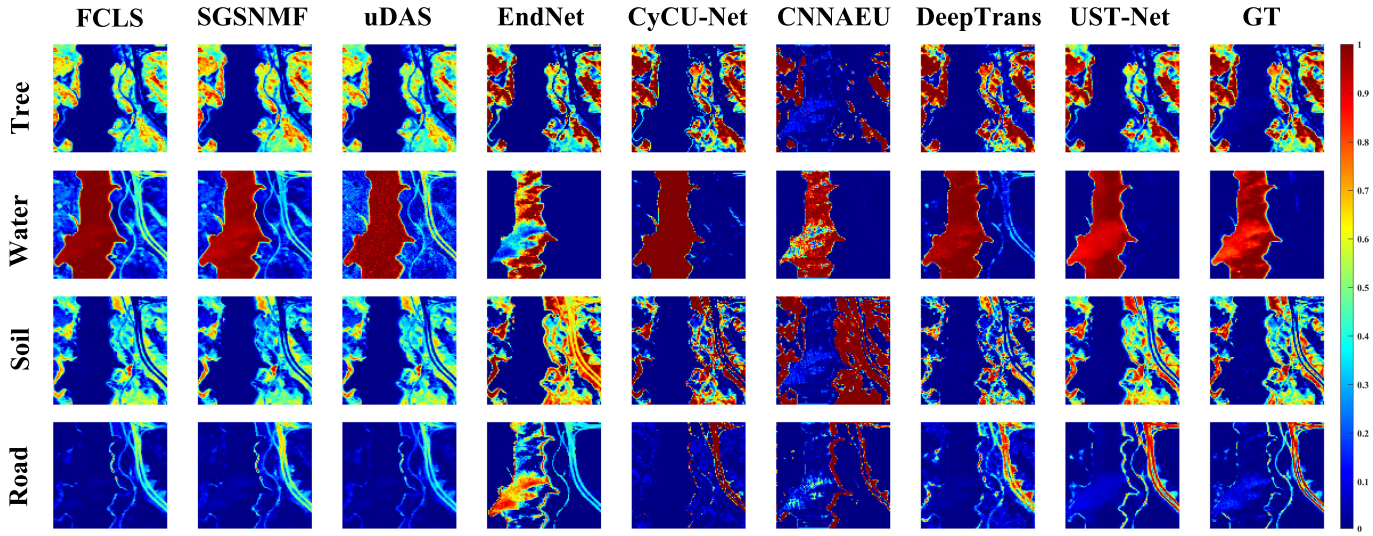


Fig. 12. Comparison of the abundance estimation results of tree, water, soil, and road with the abundance GT.

TABLE IV

QUANTITATIVE COMPARISONS ON THE WASHINGTON DC MALL DATASET. THE BEST RESULTS ARE SHOWN IN RED, AND THE SUBOPTIMAL RESULTS ARE SHOWN IN BLUE

Methods		VCA-FCLS	SGSNMF	uDAS	EndNet	CyCU-Net	CNNAEU	DeepTrans	UST-Net
SAD	Grass	0.2134	0.2133	0.2702	0.0645	0.3680	0.0749	0.2394	0.2527
	Tree	0.1152	0.2147	0.1069	0.0657	0.2342	0.1512	0.1593	0.0917
	Road	0.5682	0.3533	0.5485	0.1228	0.1698	0.4873	0.0613	0.0421
	Roof	0.0326	0.2817	0.1923	0.8509	0.4961	0.8415	0.3838	0.4911
	Water	0.0813	0.0217	0.0772	0.0250	0.1627	0.0221	0.0744	0.0304
	Trail	0.1734	0.1719	0.1414	0.0242	0.0674	0.1613	0.1105	0.0758
Mean SAD		0.1974	0.2094	0.2228	0.1922	0.2497	0.2897	0.1714	0.1640
RMSE		0.2370	0.2007	0.2471	0.2372	0.2003	0.3265	0.1337	0.1247

As the network captures HSI at a finer scale, there is a gradual enhancement in the overall unmixing performance. The performance of endmembers extraction makes a great promotion by introducing the multiscale operations. Moreover, the enhanced precision observed in the RMSE indicates the rationality of the multiscale framework within the HU. After replacing the

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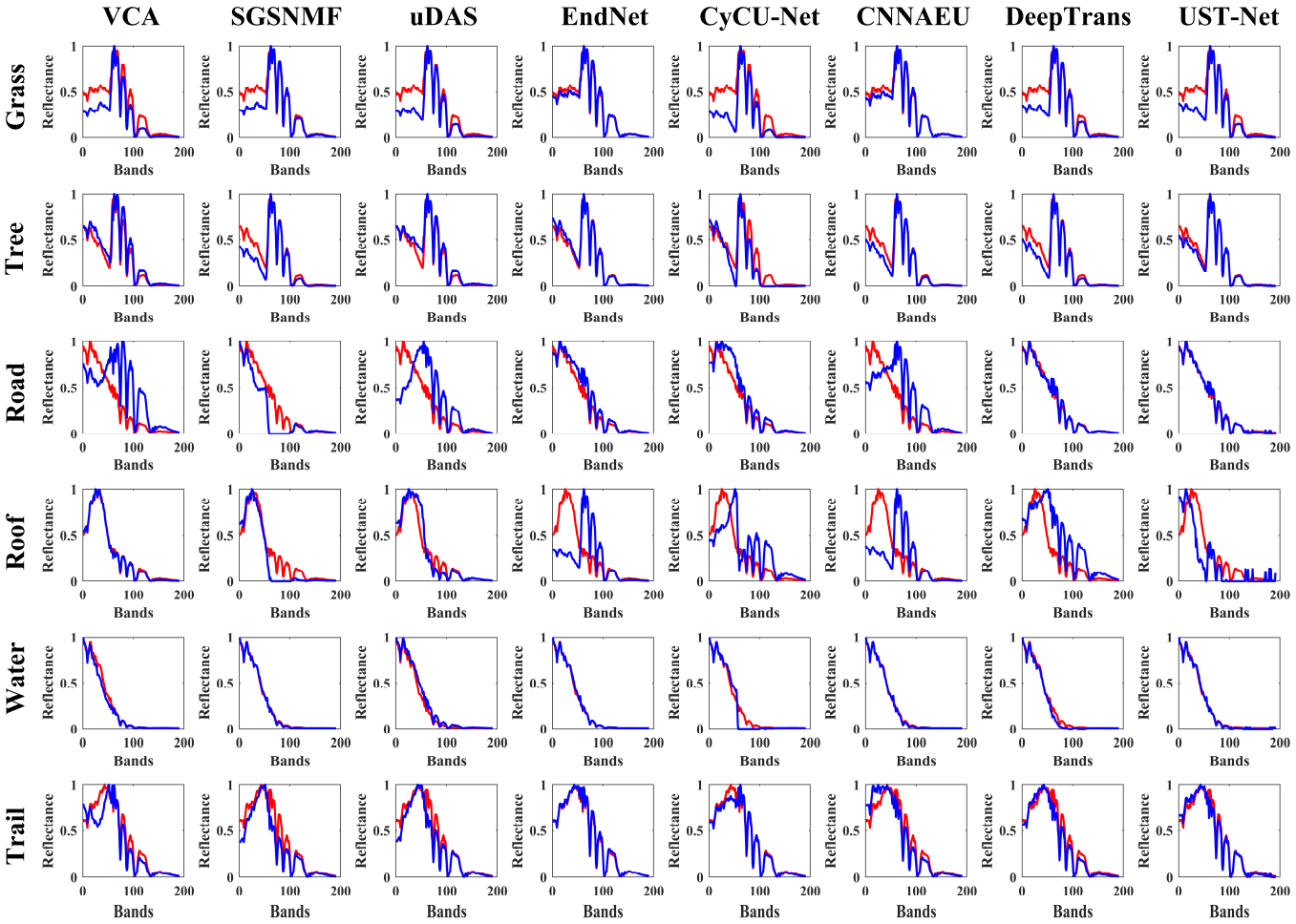


Fig. 13. Endmember extraction results of the Washington DC Mall dataset, where the red line is the true value and the blue line is the extraction results of different methods.

TABLE V
QUANTITATIVE RESULTS OF ABLATION EXPERIMENTS

	Simulated data (30 dB)		Samson		Jasper Ridge		Washington DC Mall	
	SAD	RMSE	SAD	RMSE	SAD	RMSE	SAD	RMSE
AE network based on STB with one Down-Up Block	0.0073	0.0524	0.0790	0.0974	0.0627	0.1124	0.2038	0.2382
AE network based on STB with two Down-Up Blocks	0.0057	0.0511	0.0403	0.0822	0.0588	0.1011	0.1848	0.1954
AE network based on STB with three Down-Up Blocks	**0.0030**	**0.0430**	**0.0233**	**0.0645**	**0.0432**	**0.0726**	**0.1640**	**0.1247**
AE network based on 3×3 convolution filter with three Down-Up Blocks	0.0081	0.0918	0.0400	0.0937	0.0840	0.1271	0.2226	0.3038
AE network based on 5×5 convolution filter with three Down-Up Blocks	0.0062	0.0785	0.0315	0.0726	0.0623	0.1142	0.2203	0.2663
AE network based on 7×7 convolution filter with three Down-Up Blocks	0.0072	0.0964	0.0325	0.0808	0.0532	0.1049	0.1864	0.1801

STB with convolutional layers with different filter sizes, the performance of the network degrades. For different HSIs, the optimal convolution filter size is inconsistent. For simple data, such as the simulated data and Samson dataset, the 5×5 convolutional filter size is better than others, and for complex data such as Jasper Ridge and Washington DC Mall dataset, the convolution filter size of 7×7 is better than others. However, for the AE network using STB, its performance is not limited by the scene because the attention mechanism can focus on more important features in the image and ignore some unnecessary context information.

F. Network Parameter Sensitivity Experiment

An ablation experiment is conducted to evaluate the effect of the number of heads (h) in the multihead self-attention mechanism in STB and the dimensionality (d) of low-dimensional image compression on the network's performance. The

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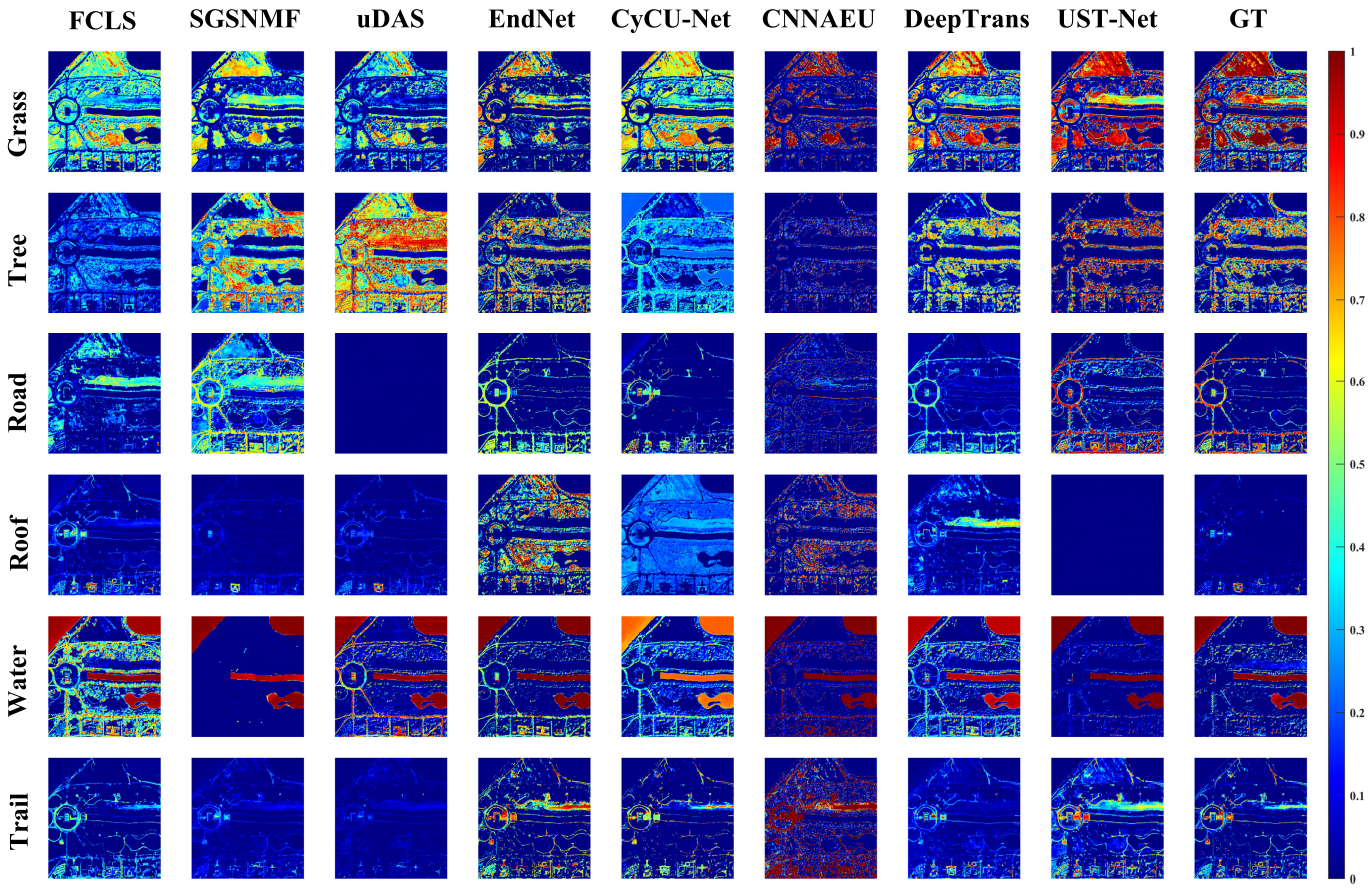


Fig. 14. Comparison of the abundance estimation results of grass, tree, road, roof, water, and trail with the abundance GT.

TABLE VI

SAD (MEAN SAD) AND RMSE OF THE SIMULATED AND REAL DATASETS WITH DIFFERENT NUMBERS OF ATTENTION HEADS AND COMPRESSED DIMENSIONS. THE BEST RESULTS ARE SHOWN IN BOLD BLACK

h	d	Simulated data (30 dB)		Samson		Jasper Ridge		Washington DC Mall	
		SAD	RMSE	SAD	RMSE	SAD	RMSE	SAD	RMSE
5	5	0.0050	0.0644	0.0281	0.0926	0.0622	0.0770	0.1540	0.1902
	10	0.1015	0.2106	0.0239	0.0840	0.2313	0.1657	0.2249	0.2983
	15	0.1012	0.2184	0.0256	0.0833	**0.0432**	**0.0726**	0.1325	0.2032
	20	0.0047	0.0583	0.0260	0.0867	0.0514	0.0667	**0.1139**	**0.1639**
	25	0.1005	0.2149	0.0431	0.0724	0.0939	0.0875	0.2137	0.2507
6	6	0.0076	0.0696	**0.0233**	**0.0645**	0.2269	0.2130	0.1852	0.2434
	12	0.0062	0.0525	0.0242	0.0721	0.0737	0.0742	0.2464	0.2379
	18	0.0051	0.0490	0.0246	0.0764	0.0492	0.0858	0.1819	0.2364
	24	**0.0049**	**0.0564**	0.0245	0.0548	0.0471	0.0632	0.2113	0.3246
	30	0.1171	0.2041	0.0791	0.1136	0.0481	0.0831	0.2054	0.2302

learning rate for all datasets is set to 0.001, and the values of γ and epoch are kept consistent with Table I.

Table VI presents the results of the experiment. The optimal number of heads and dimensions varied based on the dataset. The best results for the simulated dataset can be obtained with a number of six heads and a dimension of 24. For the Samson dataset, a head number of 6 and a dimension of 6 can achieve the best results. For the Jasper Ridge dataset, a head number of 5 and a dimension of 15 can produce balanced results. For the Washington DC Mall dataset, a head number of 5 and a dimension of 20 can result in the best SAD and RMSE. Our experiments confirm that setting the number of heads to 5 or 6 and the dimension to 1–4 times the number of heads can result in better unmixing performance and less data jitter. The

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TABLE VII
SAD (MEAN SAD) AND RMSE OF THE SIMULATED AND REAL DATASETS WITH DIFFERENT LEARNING RATES. THE BEST RESULTS ARE SHOWN IN BOLD BLACK

Learning rate	Simulated data (30 dB)		Samson		Jasper Ridge		Washington DC Mall	
	SAD	RMSE	SAD	RMSE	SAD	RMSE	SAD	RMSE
0.001	**0.0049**	**0.0564**	**0.0233**	**0.0645**	**0.0432**	**0.0726**	**0.1139**	**0.1639**
0.002	0.0983	0.2144	0.0249	0.0859	0.0509	0.0695	0.1920	0.2312
0.003	0.6760	0.3267	0.0334	0.1034	0.0768	0.1312	0.2143	0.2053
0.004	0.1132	0.1940	0.0291	0.0887	0.0581	0.0579	0.4549	0.3059
0.005	0.1347	0.1829	0.0369	0.0965	0.0589	0.0700	0.2262	0.2672
0.006	0.1260	0.1923	0.0352	0.0884	0.0560	0.0842	0.7412	0.4597

TABLE VIII
SAD (MEAN SAD) AND RMSE OF DIFFERENT LOSS FUNCTIONS ON THE SIMULATED DATASET AND REAL DATASETS. THE BEST RESULTS ARE SHOWN IN BOLD BLACK

Loss function	Simulated data (30 dB)		Samson		Jasper Ridge		Washington DC Mall	
	SAD	RMSE	SAD	RMSE	SAD	RMSE	SAD	RMSE
SAD	0.0079	0.0524	0.0277	0.0505	0.0477	0.0779	0.2086	0.2941
MSE	0.0106	0.0452	0.0540	0.2766	0.2230	0.1384	0.1944	0.3336
SAD+MSE	**0.0049**	**0.0564**	**0.0233**	**0.0645**	**0.0432**	**0.0726**	**0.1139**	**0.1639**

learning rate is tested between 0.001 and 0.006. The choices for h and d are selected based on the best results above. From Table VII, it is clear that all data achieve good results when the learning rate is set to 0.001.

G. Experiment With Different Loss Functions

To evaluate the effect of various loss functions for network training on different datasets, an ablation experiment with multiple loss functions is conducted. Table VIII shows the experimental results. The results indicate that using MSE as a loss function for network training improves the abundance extraction performance of the network on simulated data and the endmember extraction performance for the Washington DC Mall dataset. However, for the Samson, Jasper Ridge, and Washington DC Mall datasets, using MSE as a loss function results in significantly worse abundance extraction results. Using a weighted sum of SAD and MSE as the loss function on the simulated and Samson datasets can improve the accuracy of endmembers. Meanwhile, it results in improved endmember and abundance accuracy on other datasets.

V. CONCLUSION

In this article, a new HU network named UST-Net is proposed. Our network utilizes the multihead self-attention mechanism based on the shifted window to establish connections between distant pixels and obtain a global receptive field. By transferring feature information of different scales, HU performance is gradually improved, leading the network to produce higher quality abundance extraction results. The UST-Net improves the results of abundance estimation leading to positive results for endmember extraction as well. However, the current loss function for the network is a simple addition of SAD and MSE, with the latter significantly impacting network training. Therefore, an important future direction is setting learnable weight parameters for adaptive adjustment during network optimization.

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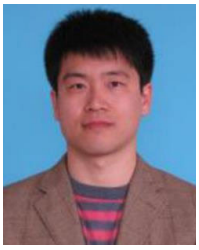
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